

Biomass-Derived Carbon Materials as an Emerging Platform for Advanced Electrochemical Sensors: Recent Advances and Future Perspectives

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Cite This: *Ind. Eng. Chem. Res.* 2023, 62, 4628–4635



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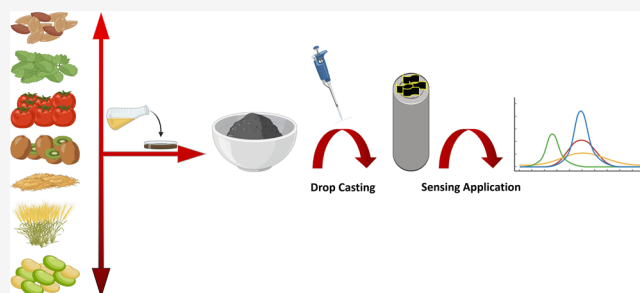


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ABSTRACT: This review describes progress in the design and synthesis of biomass-derived carbon materials with different morphologies. It also covers respective composites and their applications in the field of electrochemical sensing of (bio) chemical species. Generally, carbon-based materials have paved the way for advanced electrochemical sensing devices because of their unique physical and chemical properties. Carbon materials are typically created from chemical precursor materials; however, researchers are increasingly focusing on fabricating carbon materials from waste biomass resources due to economic and environmental constraints. Compared with other carbon materials, biomass-derived carbon materials (BCMs) are more environment-friendly with lower toxicity. In this review, we concentrated on providing researchers with a complete overview to develop an alternative strategy to improve the sensitivity of electrochemical sensors in actual samples by using numerous biomass-derived carbon-based materials modified electrodes. This review summarizes the classification, synthesis methods, and electrochemical sensing applications of BCMs and their composite materials. The major challenges in this field are outlined, and future research avenues for BCM-based electrochemical sensors are also highlighted.



1. INTRODUCTION

Nanocomposites and nanoparticles offer a wide range of possible uses, including nanoelectronics, biomedicine, solar energy, catalysts, and so on.¹ Scientists must recognize that agricultural waste biomass-assisted synthesis is a low-cost, environmentally friendly, and renewable technique, implying that agricultural wastes are appropriate renewable resources for fabricating nanomaterials to replace harmful materials.² Biobased economies and renewable biomass as raw materials have been seen as long-term solutions to local and global environmental concerns.³ Agricultural waste biomass, particularly lignocellulosic biomass (e.g., rice husk and walnut shell), has received increasing attention as an inexpensive renewable resource for producing fuels and, more recently, chemicals and materials.⁴ Biomass materials derived from organisms, including chitosan, cellulose, silk fibroin, collagen fiber, and gelatin, are abundant and renewable natural macromolecular compounds.⁵ These biomass materials possess outstanding biocompatibility and biodegradability, but their thermal stability and mechanical properties are relatively poor.⁶ Luckily, biomass materials include a lot of reactive groups, including amino-, hydroxyl-, or carboxyl groups, which can be further modified to improve their overall properties.⁷ In addition, to improve performance and safety, the main remaining challenges for the future development and widespread use of biomass materials are reduction of production

and overall device costs, development of environmentally friendly production processes, and recyclable large-scale processing units.⁸

Electroanalytical chemistry has become a research hotspot in analytical chemistry because of its inexpensive apparatus, high sensitivity, good selectivity, and rapid response.⁹ As we all know, bare electrodes often suffer from a narrow linear range, an inadequate limit of detection (LOD), and poor sensitivity.¹⁰ Therefore, effectively immobilizing electroactive species on the electrode surface has been a long-standing problem in electroanalytical chemistry.¹¹ Graphene, carbon nanotubes (CNTs), and other carbon nanomaterials have been widely used in electrode modification over the past decades, due to their unique properties such as high electrical conductivity, substantially large surface area, and strong catalytic activity.¹² However, these materials tend to form close-packed structures on the electrode surface, decreasing their specific surface area and limiting mass transfer.¹³ More recently, sustainable

Special Issue: Nanobiomaterials for Biosensing and Biodegradation

Received: August 25, 2022

Revised: December 8, 2022

Accepted: December 9, 2022

Published: December 27, 2022



biomass-derived porous carbon has received growing attention due to raw materials' low cost and easy accessibility.¹⁴ The biomass-derived porous carbon materials own more significant pore volumes and a better-regulated microstructure, which have become one of the most promising electrode materials for diverse applications such as energy storage, sensors, and wastewater treatment.¹⁵ Biomass-derived activated carbons prepared by traditional pyrolysis and hydrothermal treatment remain viable commercial alternatives because of their cheapness and large-scale production capabilities.¹⁶

There have been several comprehensive reviews summarizing the advances in the design, synthesis, characterization, electrochemical behavior, and energy storage applications of biomass-derived carbon materials (BCMs). However, there are no dedicated reviews on BCM-based electrochemical sensors, which have been fabricated by some groups over the past few years. The rapid developments in this field provide a strong impetus for writing a review on BCMs-based electrochemical sensors. Therefore, we summarize the developments in BCMs-based electrochemical sensors over the past five years.

2. CARBON-BASED MATERIALS

In the recent decade, research on carbon-based nanomaterials has been a fast-growing academic field.¹⁷ The unique characteristics and diversity of carbon-based structures and the emergence of new opportunities in many subfields of chemistry, physics, and engineering, subsequently provide the impetus for creating several sectors.¹⁸ A wide range of water pollutants, including hazardous metal ions, medicines, pesticides, metalloids, and other inorganic and organic compounds, can be adsorbed by carbon-based materials.¹⁹ No other element can produce as many different and distinct shapes and properties as carbon, such as three-dimensional (3D) diamond crystal and graphite, two-dimensional (2D) graphene, one-dimensional (1D) CNTs, and zero-dimensional (0D) fullerene molecules.²⁰ Carbon materials outperform many other materials in terms of hardness, optical qualities, heat resistance, radiation characteristics, chemical resistance, electrical conductivity, and surface and interface properties.

The synthesis of such carbon nanomaterials typically involves several processes requiring the pretreatment of pricey precursor feedstocks.²¹ Commercial and laboratory-scale manufacturing of carbon forms such as carbon nanofibers, carbon nanotubes, and fullerenes necessitates electrospinning, carbonization, and plasma-enhanced chemical vapor deposition, all of which require the use of costly equipment.²² Graphene, which has a greater specific surface area than single-walled carbon nanotubes (SWCNTs), is often produced by chemical and mechanical exfoliation stages, epitaxial growth, and chemical vapor deposition.²³ Graphene is also poorly dispersed in an aqueous solution. To form a stable and uniform suspension, the surface of graphene is often further functionalized with dispersion agents, including hydrophilic carboxyl groups, alkylamines, and amphiphilic polymers. Hence, significant efforts have been made to fabricate carbon-based nanomaterials for sensor applications from renewable sources via cost-effective methodologies.²⁴ Biomass-derived carbon materials are gaining appeal in this area due to abundant bioprecursors and facile synthesis procedures, which greatly reduce the cost of carbon-based electrochemical sensors. BCMs are generally prepared by pyrolysis and hydrothermal carbonization.

2.1. Classification of BCMs. **2.1.1. Zero-Dimensional BCMs.** 0D activated carbon nanospheres are a kind of popular carbon material on account of their large accessible area, low cost, and high charge capacity. The diameter of the biomass-based spheres ranges between 50 and 500 nm and 1–5 μm .²⁵ 0D carbons such as fullerenes and carbon quantum dots (CQDs) have been extensively utilized in various fields such as water treatment,²⁶ due to their mechanical robustness and ability to restrict volumetric changes.²⁷ Recently, 0D CQDs have been designed to address the shortcomings of inorganic quantum dots (QDs).²⁸ Because of their high carbon content, CQDs were successfully synthesized from biomass wastes (agricultural residues, food waste, municipal solid waste, etc.).²⁹ However, the process of refining them from raw biomass is time-consuming.

2.1.2. One-Dimensional BCMs. 1D BCMs are defined as biomass that has a fibrous or tubular structure; the former includes flax, ramie, and bacterial cellulose, while the latter contains cotton, kapok, and willow catkins. So far, several synthesis methods have been developed to obtain a variety of 1D BCMs. These 1D BCMs have demonstrated remarkable electrochemical performance, which is ascribed to the fact that the biomass carbon templates generally possess 1D fiber morphology in the microstructure. This feature can be readily inherited during the synthesis process. Due to the shape anisotropy and high aspect ratio of this 1D structure, the charge transportation rate can be facilitated along a radial direction, and improved interfacial polarization can be achieved as well.³⁰ On the other hand, the porous walls of carbon microtubes provide enough contact between the active material and the electrolyte, which reduces ion diffusion length and boosts electrochemical kinetics, eventually resulting in outstanding rate capability and cycle stability. Furthermore, hollow 1D carbon microtubes have great potential to be useful substrates for developing diverse redox-reaction materials.

2.1.3. Two-Dimensional BCMs. 2D carbonaceous materials with many sp^2 hybridizations have high energy storage and conversion potential because of their abundant active surface edges and strong in-plane covalent bonding. Due to their large specific surface area and good physical characteristics, 2D carbon materials have received growing attention in the field of electrochemical sensors in recent years.

2.1.4. Three-Dimensional BCMs. As is well-known, microstructures significantly impact the capacitive performances of electrode materials for electrochemical approaches; as dimensionality increases, more specific facets are exposed in the electrolyte, increasing the specific surface area and improving the capacitive capabilities. In light of this, a three-dimensional structure with well-connected tiny and large pores offers a continuous electron channel for efficient electrical contact and thus accelerates ion transfer by shortening diffusion paths; this is important for creating high-performance electrode materials. As a result, 3D microstructures open up new possibilities for the rational design and synthesis of innovative architectures tailored to improve electrochemical performance.

2.2. Synthesis Methods for BCMs. **2.2.1. Biomass Pyrolysis.** Pyrolysis, also referred to as thermochemical conversion, is often used to produce carbon materials from biomass. It is usually done using a flowing gas in the absence of oxygen at high temperatures. Carbon backbones from biomass sources can form amorphous or partially crystalline carbon depending on the temperature, temperature ramping rate,

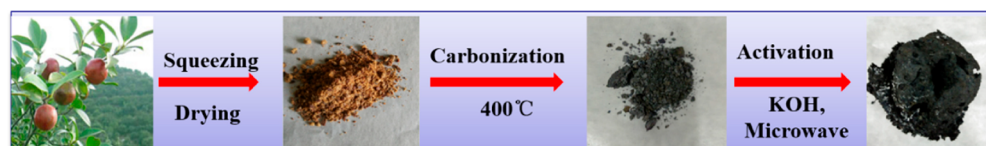


Figure 1. Schematic diagram for the fabrication of mesoporous carbon nanosheets. Reproduced with permission from ref 42. Copyright 2019 Elsevier.

particle size, and catalyst used in pyrolysis. In general, a low heating rate favors char formation, and a high heating rate facilitates the formation of volatile compounds.

2.2.2. Hydrothermal Carbonization. Hydrothermal carbonization (HTC) is an innovative, low-cost, environmentally friendly technique to fabricate functional carbon compounds using carbon-rich precursors. HTC is typically performed at low processing temperatures ($<300\text{ }^{\circ}\text{C}$), with water as the carbonization medium, and self-generated pressures. It is feasible to modify the final morphology, porosity, and physicochemical characteristics using colloid and polymer science methods to fulfill the intended applications.³¹ During HTC, soft biomass with less crystalline cellulose and low molecular weight compounds were first broken down, followed by the formation of certain globular biomass-derived carbon composites with very tiny diameters and interstitial holes. Carbonization temperature and residue time are two critical aspects that directly determine whether HTC will succeed or not.³²

2.2.3. Physical and Chemical Activation. Activation is the most practical method for increasing the surface area of BCMs and adjusting the meso/micropore balance. The activation strategies may be classified into three types based on the activation mechanism: chemical activation, physical activation, and self-activation.³³ However, contrary to chemical and physical methods, no additional activation materials are needed for a self-activation approach, simplifying this method and decreasing the cost.

The activation of carbon materials with gaseous agents such as O_2 , CO_2 , or hot steam to form porous structures is commonly referred to as physical activation. On the other hand, chemical activation is related to specific chemical activators (such as acid, alkali, or salt) functioning at high temperatures ranging from 450 to $900\text{ }^{\circ}\text{C}$.³⁴ Therefore, different activation procedures mainly control pore distributions and electrochemical performances. However, high-temperature activated carbon materials have superior electronic conductivity; nevertheless, high temperatures in steam-activated activated carbon materials can degrade the biomass precursor, reducing surface areas and pore content in the resultant activated carbon materials. As a result, chemical activation is more suitable for effective pore dispersion control to avoid the collapse of pure biomass precursor structures.³⁵

2.2.4. Template-Assisted Method. For the preparation of graphene, natural 2D templates, metal oxides, and metal salts have been employed as templates. These templates can be removed by acid- or water-etching to generate high-quality graphene.³⁶ Metal salts like FeCl_3 and NaCl are frequently employed as templates to make graphene and graphene-like carbons.³⁷ Metal salts with a layered structure may have a small gap between them. In addition to a single salt such as NaCl , molten salts such as KCl/KNO_3 are also employed as a template for synthesizing G-carbons.³⁸ At high temperatures, molten salt is the equivalent of a liquid solvent, capable of

dissolving large amounts of raw materials and recrystallizing the carbon source into its unique layered structure after cooling. Nitrate combines with carbon to generate CO_2 , N_2 , NO , and other gases at high temperatures, resulting in a greater number of micropores and a larger specific surface area than single salts.

2.2.5. Additive-Assisted Method. Although templating methods (including soft-templating and hard-templating) effectively produce controlled morphologies and porous characteristics, they are costly and incompatible with heteroatom doping.³¹ Additive-assisted pyrolysis combines metal salts such as sodium, potassium, and calcium salts and inorganic additives (i.e., zeolites and charcoal), offering advantages over conventional pyrolysis. It may lower the required operating temperature, cracking time, and solid residue yield and increase waste cracking efficacy and pyrolysis product quality.³⁹

2.2.6. Microwave-Assisted Heating. Microwaves are electromagnetic waves with wavelengths ranging from 1.0 to 1.0 mm and frequency ranging from 300 MHz to 300 GHz . Microwaves were initially utilized in telecommunications, such as radar and telephone technologies.⁴⁰ Compared to conventional heating, microwave heating is accelerated by dipole rotation and internal ion conduction, and the direction of heat transfer is opposite to that of regular heating. Consequently, microwave heating solves the shortcomings of traditional heating, such as increased heat loss, low energy efficiency, and high cost. More importantly, selective heating characteristics of microwave heating can enhance chemical reactions and improve product quality.⁴¹ As a result, microwave-assisted heating has recently gained popularity for biomass conversion and material synthesis. For example, Bo et al. used a microwave treatment to prepare a mesoporous carbon nanosheet with rich pseudocapacitive groups from camellia oleifera biomass waste (Figure 1).⁴² This method resulted in mesoporous carbon nanosheets with a large surface area and a high oxygen content. Furthermore, when compared to the usual activation techniques developed from DAC-800, the resulting mesoporous carbon nanosheets demonstrated a high specific capacitance and good rate performance.

2.2.7. Ionothermal Carbonization. Another effective technique for the synthesis of BCMs is ionothermal carbonization, which allows for low-cost biomass conversion to useful carbonaceous materials at relatively mild settings (180 – $220\text{ }^{\circ}\text{C}$).⁴³ Moreover, numerous heteroatoms may be easily doped into the final carbon framework.⁴⁴ The ionothermal method for converting biomass into carbons allows for the retention of a high level of oxygen and nitrogen in the resultant materials. Various forms of biomass, such as glucose, starch, lignocellulose, bamboo, Moringa oleifera branches, and so on, have been effectively employed as precursors for functional carbons using the ionothermal approach.⁴⁵ Pambel et al. recently proposed an ionothermal synthesis technique to fabricate a series of glucose-derived carbons with a tailored pore structure.⁴⁶ The

as-obtained BCMs have significant total pore volumes ($2160 \text{ m}^2 \text{ g}^{-1}$ and $1.74 \text{ cm}^3 \text{ g}^{-1}$). Xie et al. also disclosed the ionothermal synthesis of carbon-based composites from fructose in the iron-containing ionic liquid 1-butyl-3-methylimidazolium tetrachloridoferrate(III), $[\text{Bmim}][\text{FeCl}_4]$ acting as a solvent, catalyst, and template for product production.⁴⁷ The composites' mesoporosity, high magnetic moment, and huge specific surface area make them suitable for magnetic separation water purification. The proposed process proved to be flexible and scalable, paving the way for ionothermal synthesis on a larger scale to manufacture materials that, while not produced sustainably, are helpful for water treatment, such as organic molecule removal.

3. BCMs FOR ELECTROCHEMICAL SENSING APPLICATIONS

The development of BCMs has been a recent trend in the design of electrochemical sensors. As shown in Table S1, various BCMs have been used for this purpose and applied to the detection of various analytes, with a special emphasis on the detection and quantification of organic molecules, which are commonly studied by electroanalytical groups worldwide using other carbon-based electrodes as a proof-of-concept. We believe this is due to the closeness between the physicochemical and electrocatalytic characteristics of BCMs and the well-known electrode utilized in research on these molecules. Furthermore, some strategies for obtaining carbon materials with desired properties for use in electroanalysis have been employed, resulting in carbon materials with high electrical conductivity and electrochemical activity due to the spatially ordered porous structure, high graphitization, and doped nitrogen, making BCMs appealing for the preparation of electrochemical sensors. Indeed, the pyrolysis process is critical for producing high-performance BCMs, and it is employed for all the activities stated in Table S1.

Kim et al. proposed a two-step activation of BCMs, including the initial activation of kelp powder with ZnCl_2 followed by an activation step with KOH (Figure 2), for the

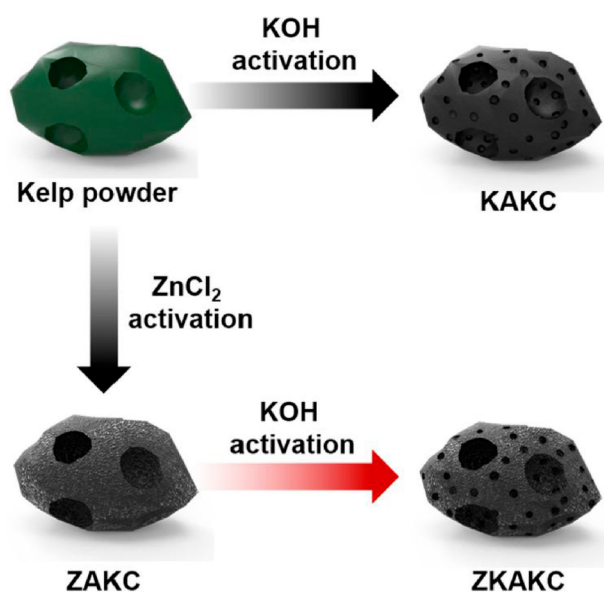


Figure 2. Graphical illustration of the preparation of ZnCl_2 - KOH activated kelp carbon (ZKAKC). Reproduced with permission from ref 48. Copyright 2018 Elsevier.

electrochemical determination of acetaminophen.⁴⁸ The resulting BCMs significantly enhanced the electroactive surface area (from 0.061 to 0.122 cm^2) and improved the electrochemical activity for acetaminophen determination. The as-synthesized materials were modified on a glassy carbon electrode (GCE) for acetaminophen determination in the presence and absence of ascorbic acid and dopamine through a differential pulse voltammetry (DPV) technique in 0.1 M phosphate buffer saline (PBS, $\text{pH} = 7.4$). The proposed sensor exhibited remarkable sensitivity toward acetaminophen determination with a rather wide linear detection range (0.02 – $20 \text{ }\mu\text{M}$) and a low LOD ($0.004 \text{ }\mu\text{M}$). To our surprise, the LOD for acetaminophen in the presence of dopamine and ascorbic ($0.007 \text{ }\mu\text{M}$) is comparable to that in PBS buffer, suggesting good selectivity.

Chen et al. combined CuCo_2O_4 /biomass-derived carbon ($\text{CuCo}_2\text{O}_4@\text{BC}$) with molecularly imprinted polymer (MIP) to fabricate a sensitive electrochemical sensor for detecting tryptophan (Trp) in real samples. The BC and CuCo_2O_4 were synthesized via pyrolyzing leaves of *Lactuca sativa* L. var. *ramosa* Hort at $800 \text{ }^\circ\text{C}$ and hydrothermal method, respectively (Figure 3).⁴⁹ The MIP film was electropolymerized on the

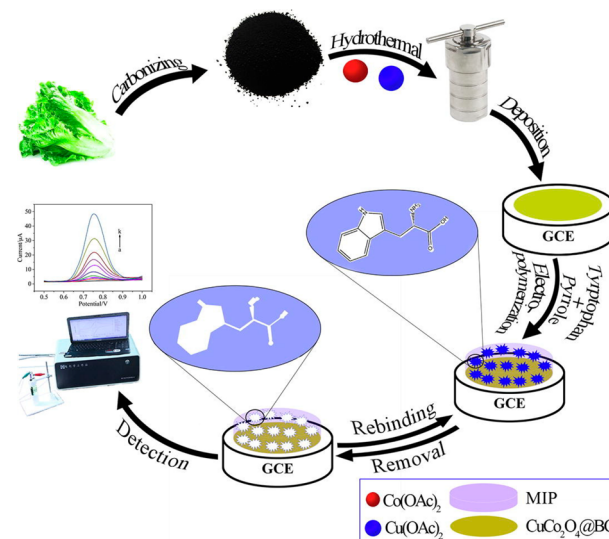


Figure 3. Demonstration of the MIP/ $\text{CuCo}_2\text{O}_4@\text{BC}$ /GCE construction process and electrochemical determination of Trp. Reproduced with permission from ref 49. Copyright 2021 Elsevier.

surface of $\text{CuCo}_2\text{O}_4@\text{BC}$ decorated GCE using Trp and pyrrole as the template and monomer, respectively, forming the specific recognition sites for binding Trp. The surface area of the developed electrode was 0.17 cm^2 , which is 2.4 times higher than that of bare GCE (0.07 cm^2). Accordingly, the MIP/ $\text{CuCo}_2\text{O}_4@\text{BC}$ /GCE showed superior electrocatalytic activity for Trp, with two linear ranges from 0.01 to $1.0 \text{ }\mu\text{M}$ and from 1.0 to $40 \text{ }\mu\text{M}$ and a low LOD of $0.003 \text{ }\mu\text{M}$. Moreover, the selectivity of MIP/ $\text{CuCo}_2\text{O}_4@\text{BC}$ /GCE was better than that of NIP/ $\text{CuCo}_2\text{O}_4@\text{BC}$ /GCE due to the specific recognition capacity of the MIP. The proposed MIP/ $\text{CuCo}_2\text{O}_4@\text{BC}$ /GCE could reliably detect Trp human serum and urine as real samples with satisfactory recovery. The results obtained by this sensor were also compared to high-performance liquid chromatography (HPLC), demonstrating promising prospects in the selective and sensitive determination of Trp in real samples.

Nikhil et al. developed a low-cost and environment-friendly 2D activated carbon (AC) modified GCE (AC-GCE) for electrochemical determination of roxarsone (ROX), an arsenic-based medicine.⁵⁰ Meso/microporous 2D-AC was prepared from a natural biomass *Desmostachya bipinnata* through chemical activation at 700.0 and 800 °C in the presence of KOH (Figure 4). The as-obtained 2D-AC significantly boosted

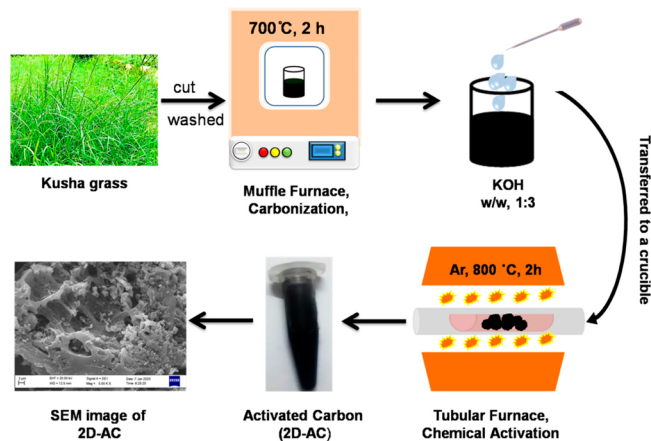


Figure 4. Synthesis procedure of 2D activated carbon using biomass. Reproduced with permission from ref 50. Copyright 2022 ACS.

the electrochemical response of ROX, mainly due to the large surface area (0.075 cm²), which was about 1.44 times higher than that of bare electrode, and faster electron transfer of 2D-AC. As a result, the AC-GCE showed excellent sensing performance toward ROX, with a wide linear response range of 0.76–474 μM and an LOD as low as 1.5 nM. Moreover, the proposed AC-GCE works well in actual samples and on disposable screen-printed carbon electrodes (SPCEs), demonstrating tremendous application prospects in clinical diagnosis.

Niu et al. presented the preparation of biomass-derived porous carbon (BPC) with rich 3D interconnected frameworks by calcinating banyan leaves and its application for the voltammetric determination of luteolin.⁵¹ Au nanoflakes (AuNFs) were further decorated on BPC through the hydrothermal approach to enhance electrochemical responses. As shown in Figure 5a, Nafion-AuNFs-BPC/GCE exhibited sharper oxidation and reduction peaks than bare and other modified electrodes, indicating higher electroactivity of the developed electrode. The effective area of Nafion-AuNFs-BPC/GCE was 0.036 cm², which was 1.54 times higher than that of bare electrode. Moreover, the peak current increased with increase in the luteolin concentration and showed two linear ranges from 0.15 to 1.8 μM and from 1.8 to 10.0 μM with a low LOD of 0.07 μM for luteolin determination (Figure 5b,c). Furthermore, the Duyi Wei capsule and human urine were chosen as the real samples to assess the practical application of Nafion-AuNFs-BPC/GCE. As shown in Table S1, the recoveries are 97.0%–105% for the capsule samples and 98.5%–106% for the human urine sample, with a low relative standard deviation (RSD) of 5.00%, indicating excellent reliability and practicability of the proposed sensor.

Ganesan et al. proposed the novel and green keratinous sludge biomass-derived carbon-based molybdenum oxide (KSC@MoO₃) nanocomposite material through two-step (hydrothermal and copolyolysis) procedures for a simultaneous determination of hydroquinone (HQ) and catechol (CC).⁵²

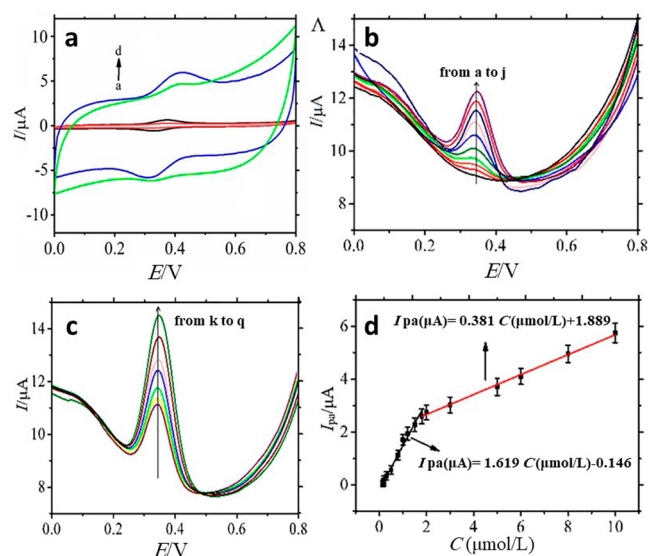


Figure 5. (a) CV curves for 5.0 μM luteolin at Nafion/GCE (curve a), GCE (curve b), Nafion-BPC/GCE (curve c), and Nafion-AuNFs-BPC/GCE (curve d) in 0.1 M PBS (pH = 6.0); DPVs recorded at Nafion-AuNFs-BPC/GCE in the presence of luteolin solutions at various concentrations (b) a–j: 0.0 to 1.8 μM; (c) k–q: 1.8 to 10.0 μM; (d) the corresponding calibration plot for luteolin. Reproduced with permission from ref 51. Copyright 2021 Elsevier.

The valorization of chicken feather waste into lipids and keratinous sludge biomass was examined, revealing the efficient and sustainable conversion of chicken feather waste into high-value lipids for biofuel and nanomaterials for fabricating a low-cost electrochemical sensor (Figure 6). The MoO₃@KSC/SPE showed sharp and well-defined anodic peaks with the highest anodic peak current while the peak-to-peak separation was remarkably decreased to 0.120 V (CC) and 0.141 V (HQ), demonstrating outstanding electrocatalytic activity toward electrocatalytic oxidation of CC and HQ. Furthermore, with a linear concentration range of 5.0–176.8 M and low LODs of 0.063 and 0.059 μM for HQ and CC, the fabricated electrode demonstrated exceptional electrochemical activity to determine HQ and CC, both independently and simultaneously. As a result, this newly designed electrode has the potential to be an outstanding sensor for practical use in the simultaneous assessment of HQ and CC in environmental water samples.

4. CONCLUSION AND OUTLOOK

Biomass-derived carbon materials have emerged as attractive materials for direct electrochemical sensing as well as robust matrices or templates for synthesizing carbon-based materials for electrochemical sensing. The most current methodologies and advancements in developing biomass-derived carbon compounds for electrochemical sensing are discussed in this review. The composites based on biomass-derived carbon components primarily concern improved electron conductivity, specific surface area, and electrochemical sensing capabilities. Biomass-derived carbon-based composites are suited for electrochemical sensing because of their long-term stability, environmental friendliness, sustainability, porous architectures, high conductivity, hydrophobicity, and ease of modification. On the other hand, other materials employed in electrochemical sensors have several disadvantages compared to biomass-derived materials, including the unavoidable inadequacies of time-consuming, high-cost, or/and complicated

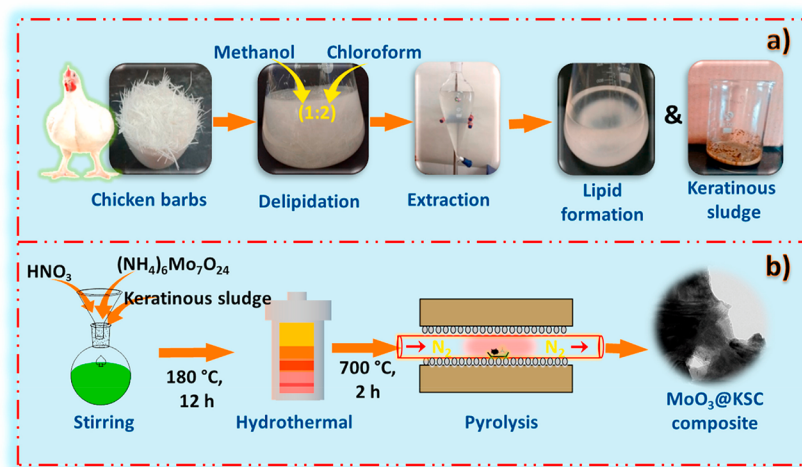


Figure 6. (a) Isolation and separation of keratinous sludge from chicken feather waste and (b) synthesis of $\text{MoO}_3\text{@KSC}$ composite. Reproduced with permission from ref 52. Copyright 2022 Elsevier.

procedures. Although substantial progress has been achieved in developing biomass-derived carbon compounds for electrochemical sensors, several obstacles remain that restrict their future applicability.

Most studies focus on nanostructured BCMs as a renewable alternative to address environmental issues, which may be attributed to their larger surface area and, in some cases, to their smaller pore sizes.⁵³ Hence, using nanosized BCMs may be an interesting strategy for developing cost-effective and highly efficient electrochemical sensors. Another feasible strategy is the graphene-like material obtained through the exfoliation of graphitized BCMs. The synthesis of graphene-like BCMs or their derivatives from biowastes has also been studied in more recent years. Graphene from BCMs may possess inherent properties of graphene, including strong mechanical strength, high electrical conductivity, large active surface area, and favorable catalytic activity. Considering their cost-effectiveness, eco-friendliness, and inherent properties, graphene-like BCMs are expected to be an attractive alternative for designing and fabricating new electrochemical sensors. As a result, the green, cost-effective, and renewable preparation of BCMs will be a talented material for the manufacture of advanced sensors and biosensors.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.2c03058>.

Applications of biomass-derived carbon materials-based electrochemical sensors for various analyte determinations (PDF)

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Notes

The authors declare no competing financial interest.

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